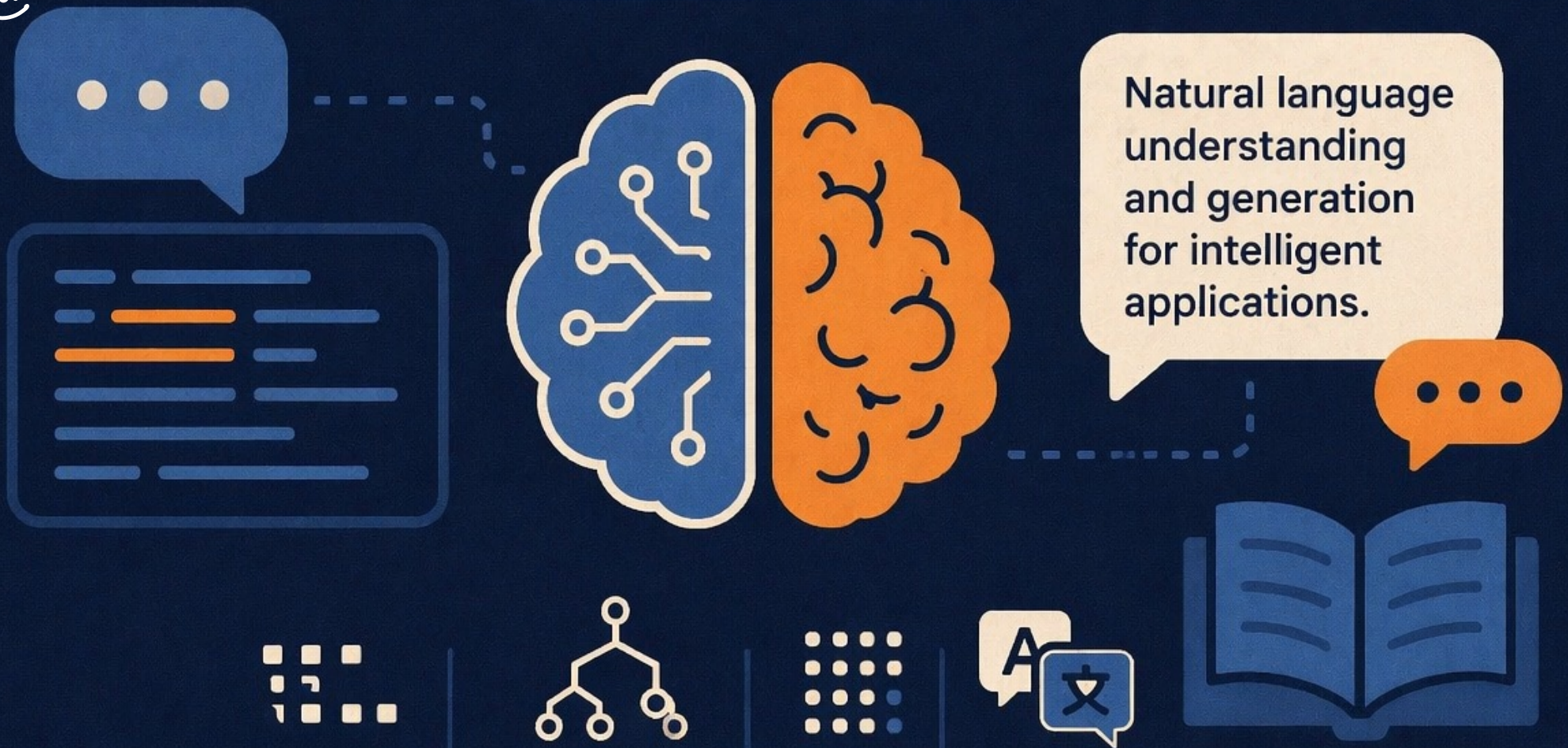
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: Transformer Continued

#Agenda

1) Understanding Decoder

2) Transformer Coding

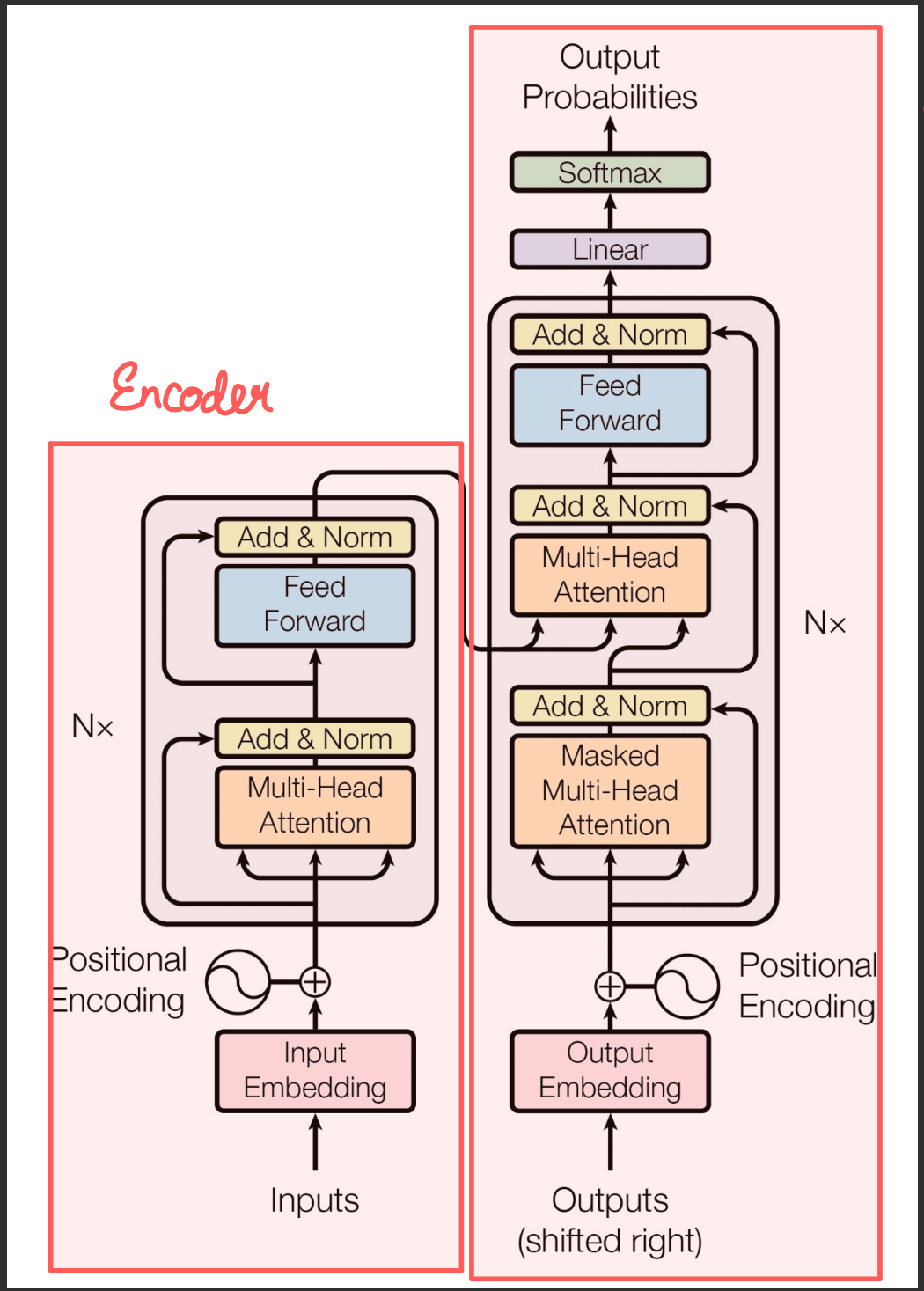
Recap

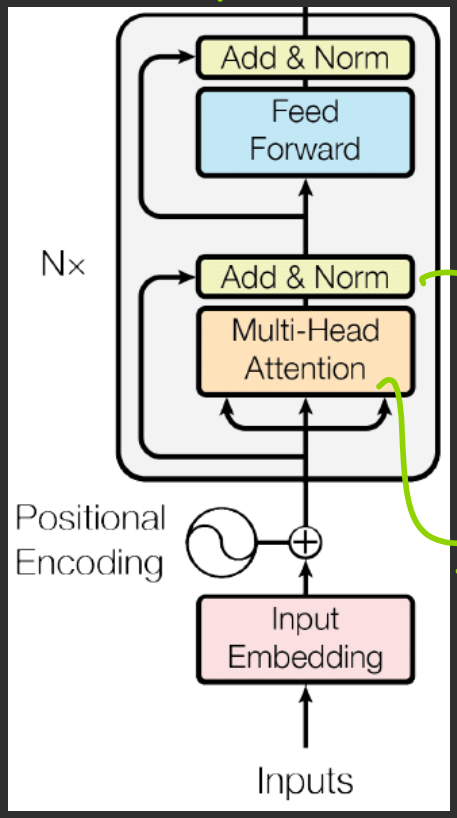
1) encoder - decoder architecture

Machine translation problem → Seq2Seq

English → Hindi

(We are friends) → (Hum dost Hain)


Output Probabilities
Softmax
Linear
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Add & Norm
Masked Multi-Head Attention
Nx
Positional Encoding
Output Embedding
Outputs (shifted right)
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Nx
Positional Encoding
Input Embedding
Inputs
Encoder
decoder



decoder

positional Encoding – It helps to track the order of words

WE + POS E = concatatenation

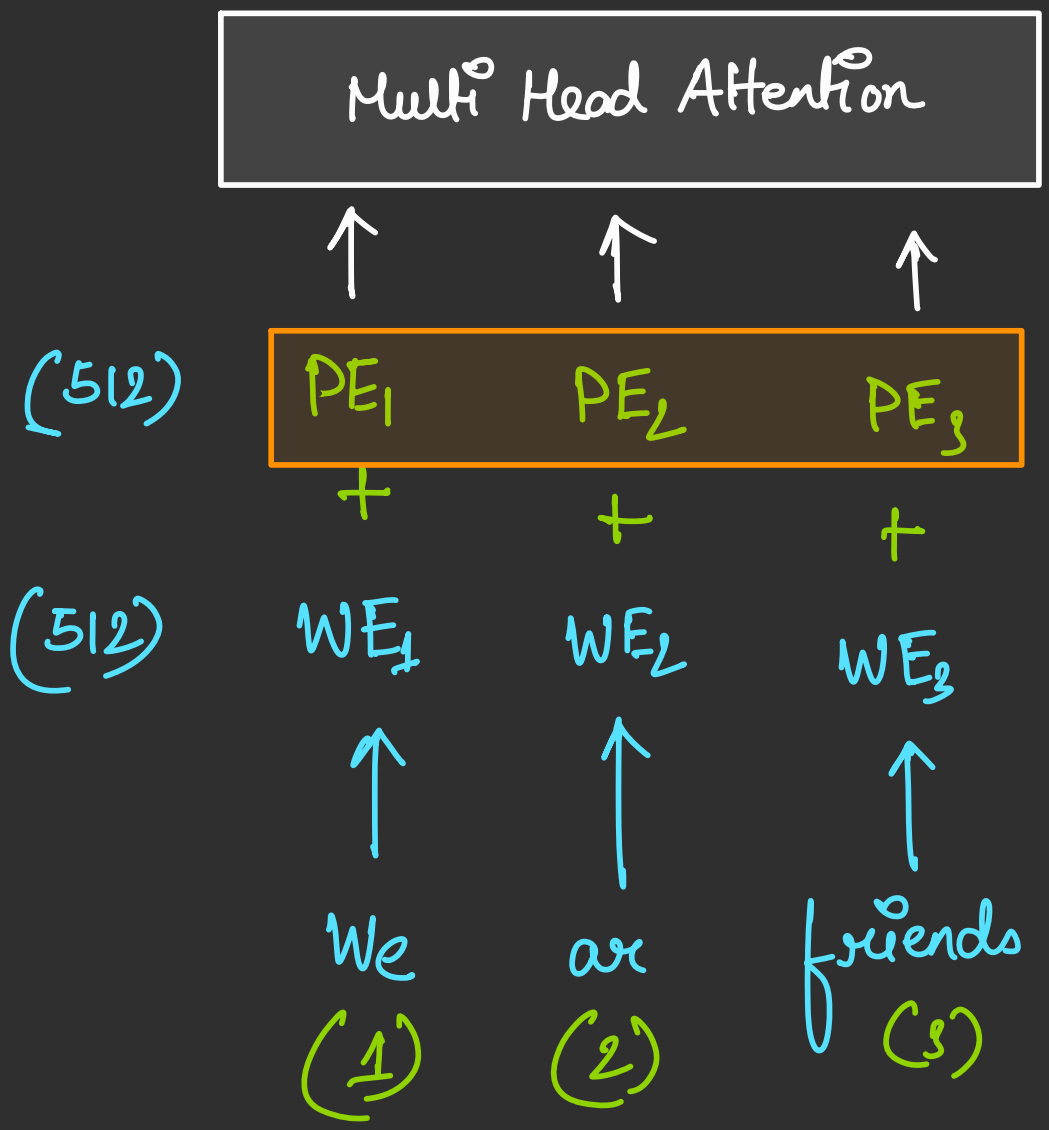
Layer

1 Block → Encoding block

N=1

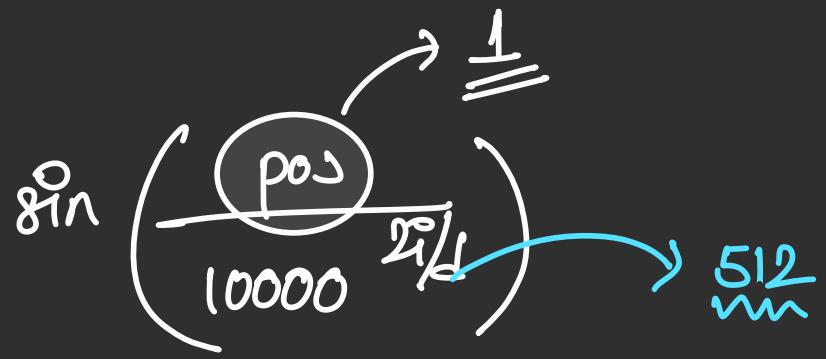
sin & cos

$$\sin\left(\frac{pos}{10000^{2i/d}}\right)$$


Multi Head Attention
(512)
PE1
PE2
PE3
+
+
+
(512)
WE1
WE2
WE3
We
are
friends
(1)
(2)
(3)

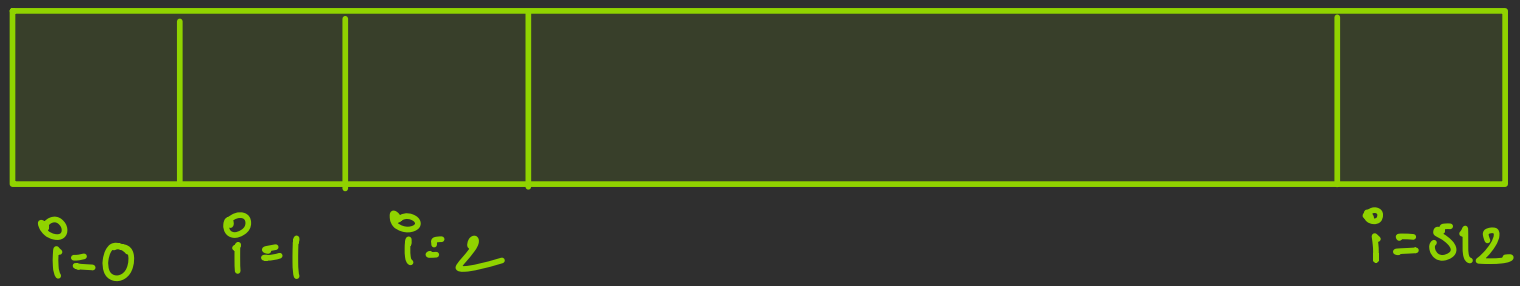
Positional Encoding

WE → 512

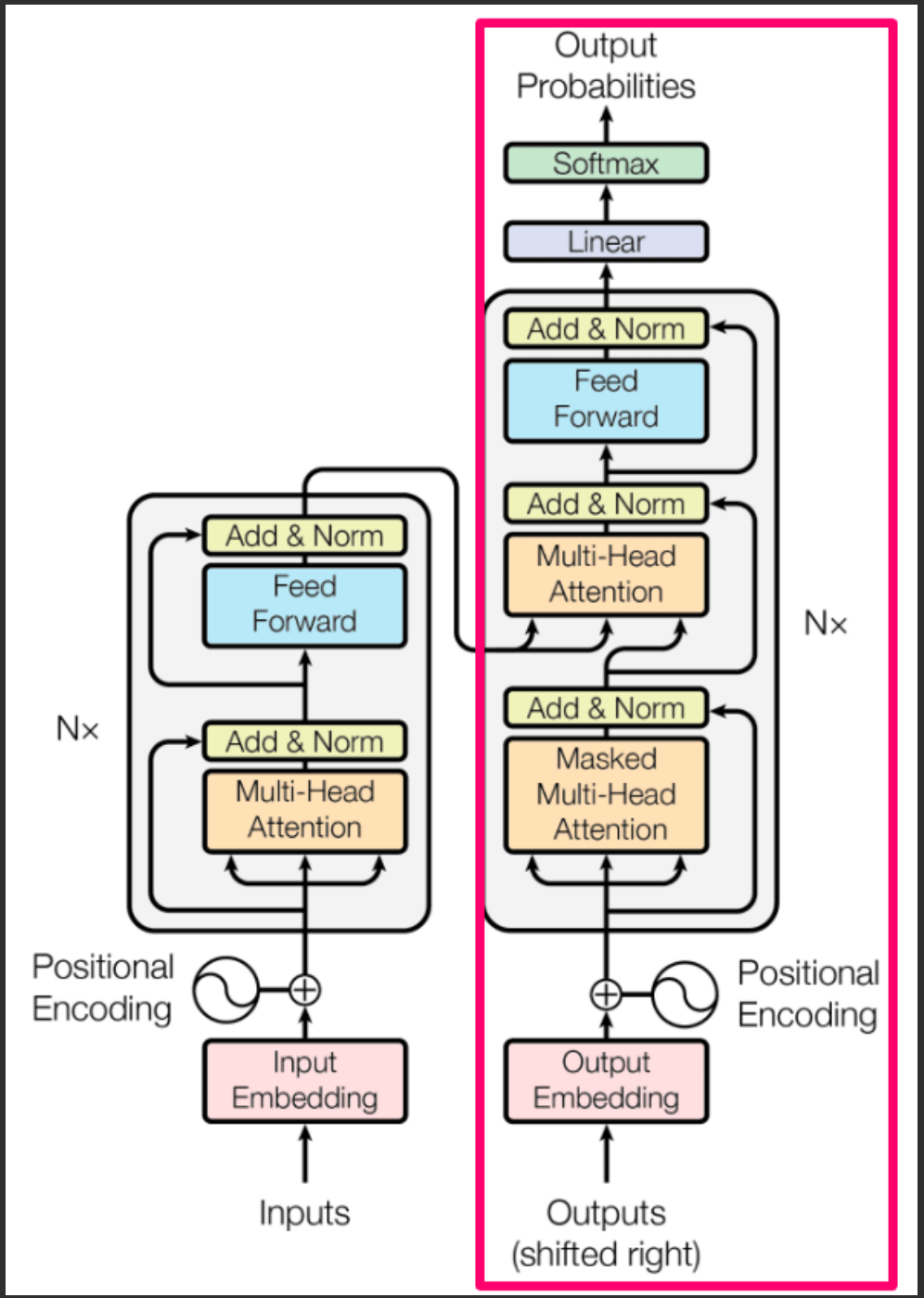


We are friend

(1)



$$\cos(x+\text{pos}) = \sin x \cdot \cos(\text{pos}) + \cos(x)\sin(\text{pos})$$



Encoder → We are friends

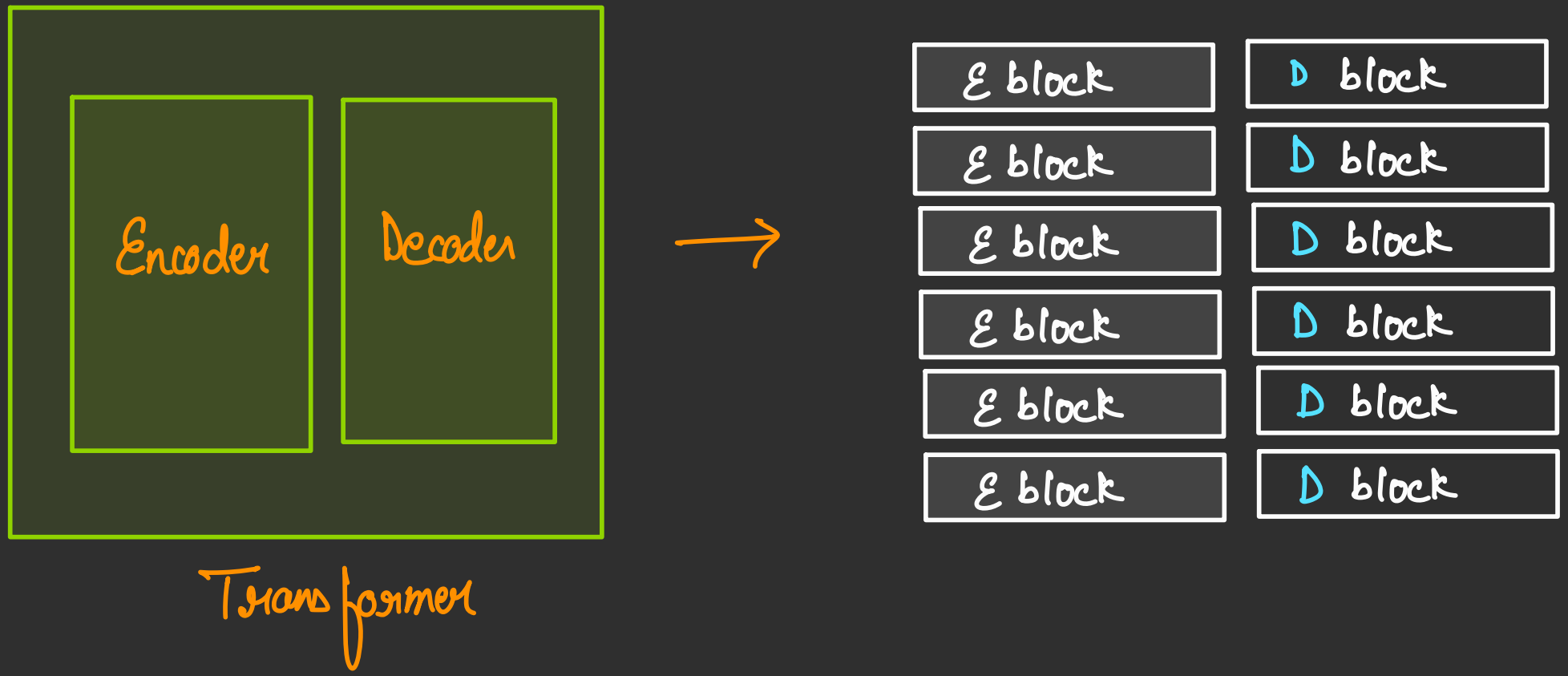
Encoder o/p → Contextual Vectors

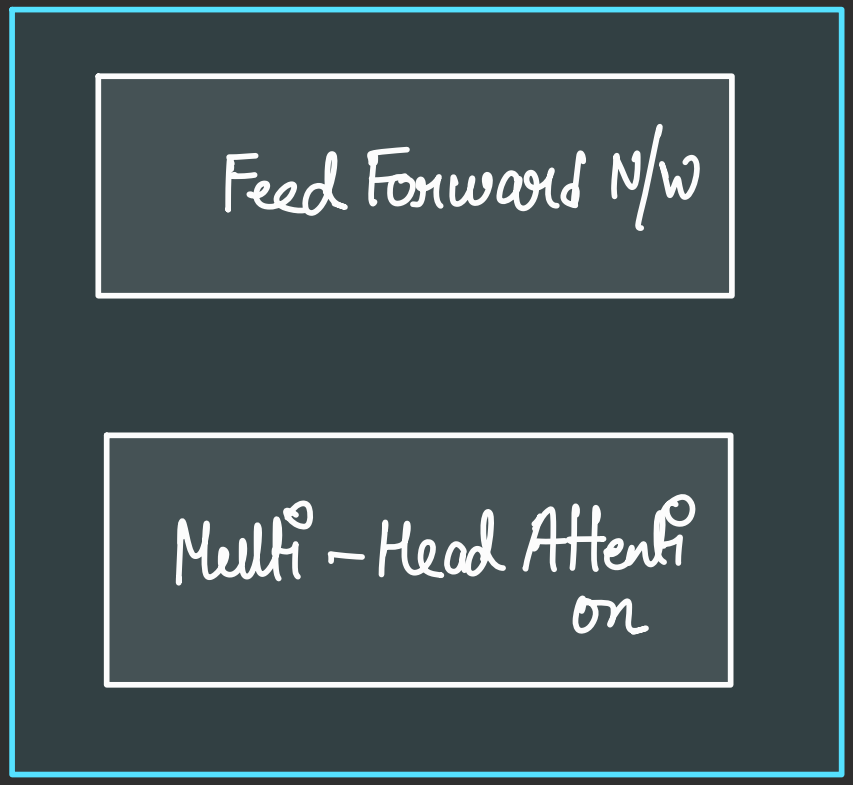
Train

Decoder → Hum dost Hain

Decoder (shift) → <start> Hum dost Hain

$WE_1 + PE_1$ $WE_2 + PE_2$ $WE_3 + PE_3$ $WE_4 + PE_4$


Encoder
Decoder
Transformer
E block
E block
E block
E block
E block
E block
D block
D block
D block
D block
D block
D block

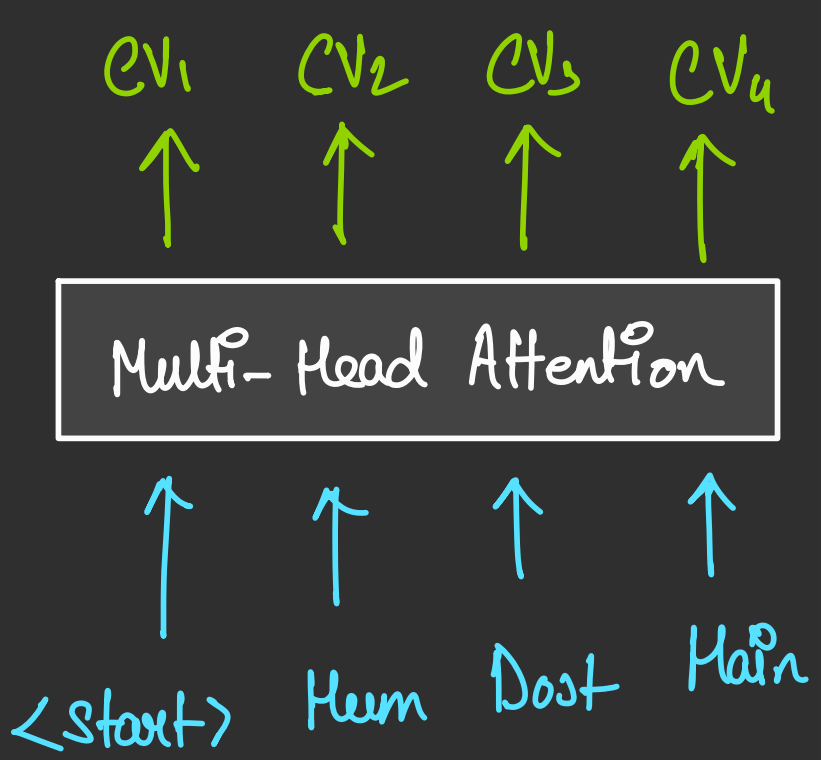


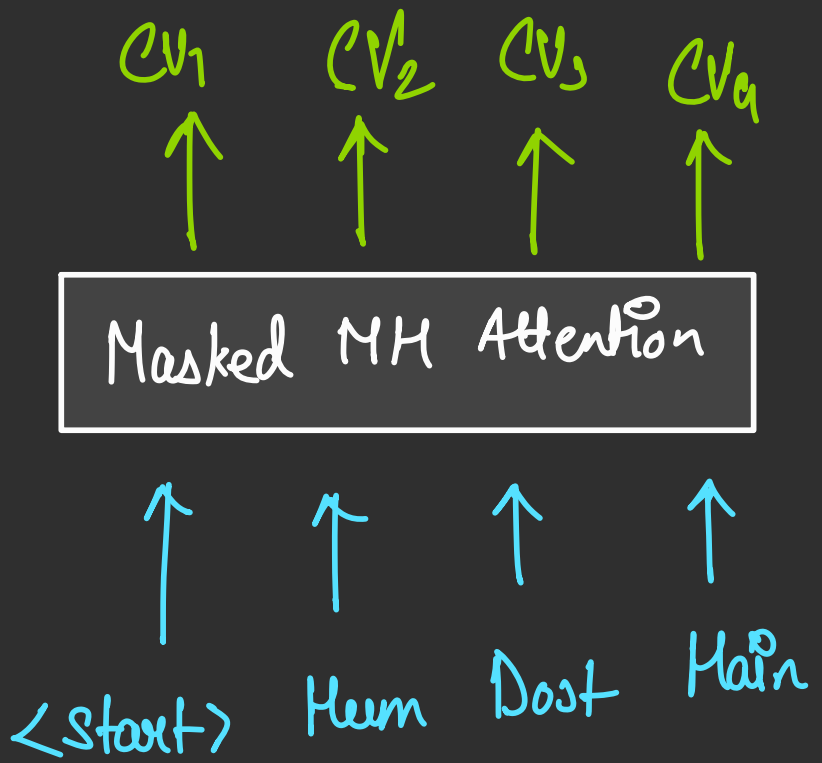
Encoder Block



Decoder Block

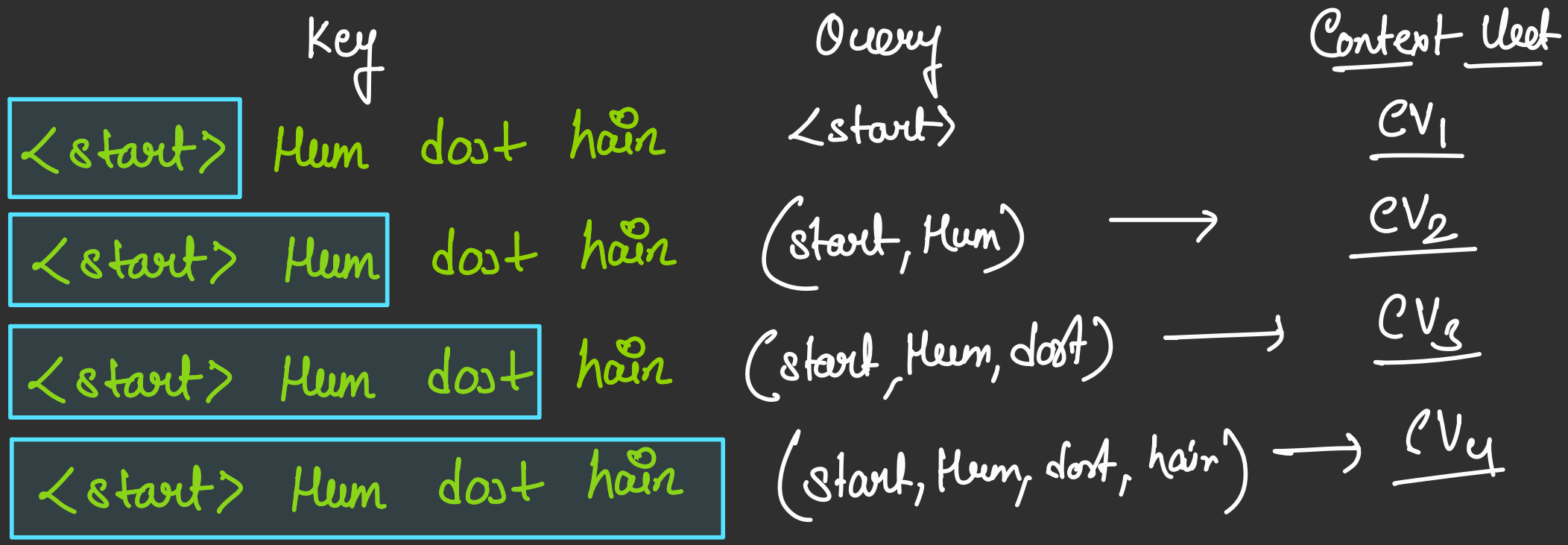
Masked Multi-Head Attention

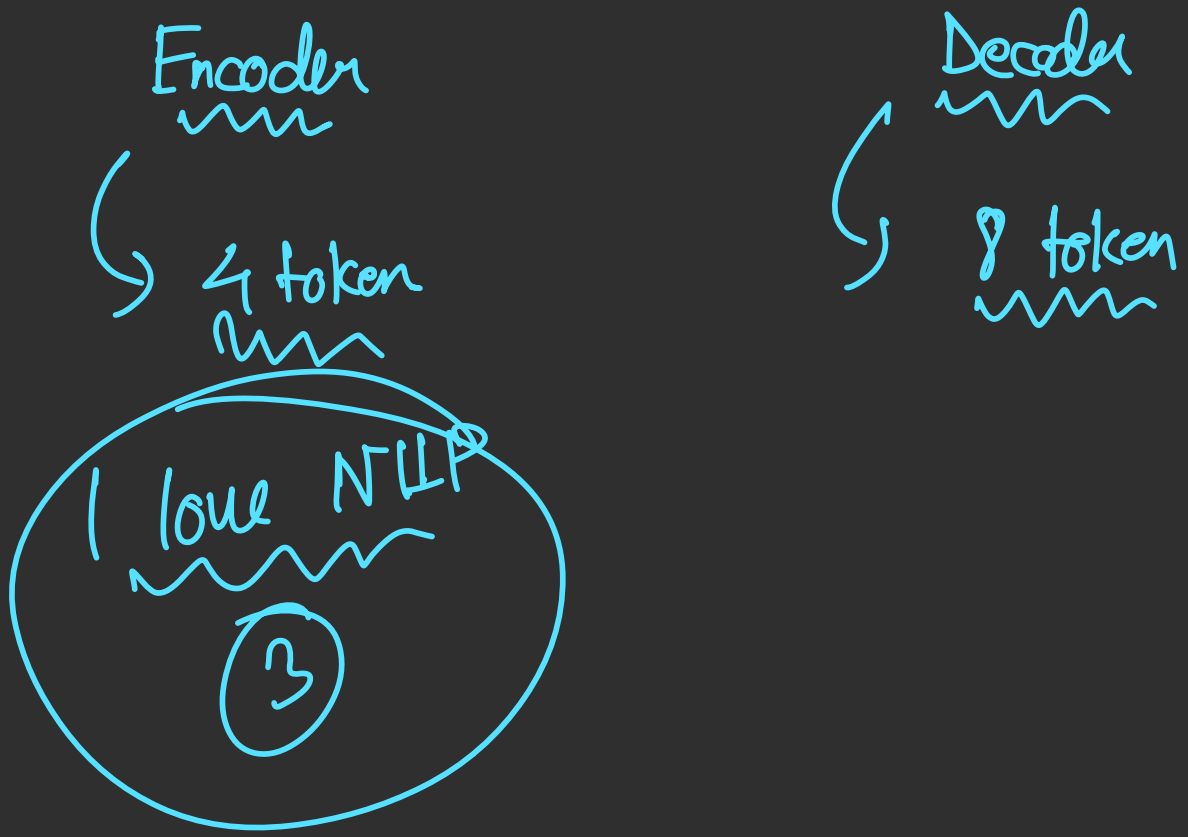




We are friends

Hum dost hain


Key
Query
Context Vect
<start> Hum dost hain
<start>
cv1
<start> Hum dost hain
(start, Hum)
cv2
<start> Hum dost hain
(start, Hum, dost)
cv3
<start> Hum dost hain
(start, Hum, dost, hain)
cv4

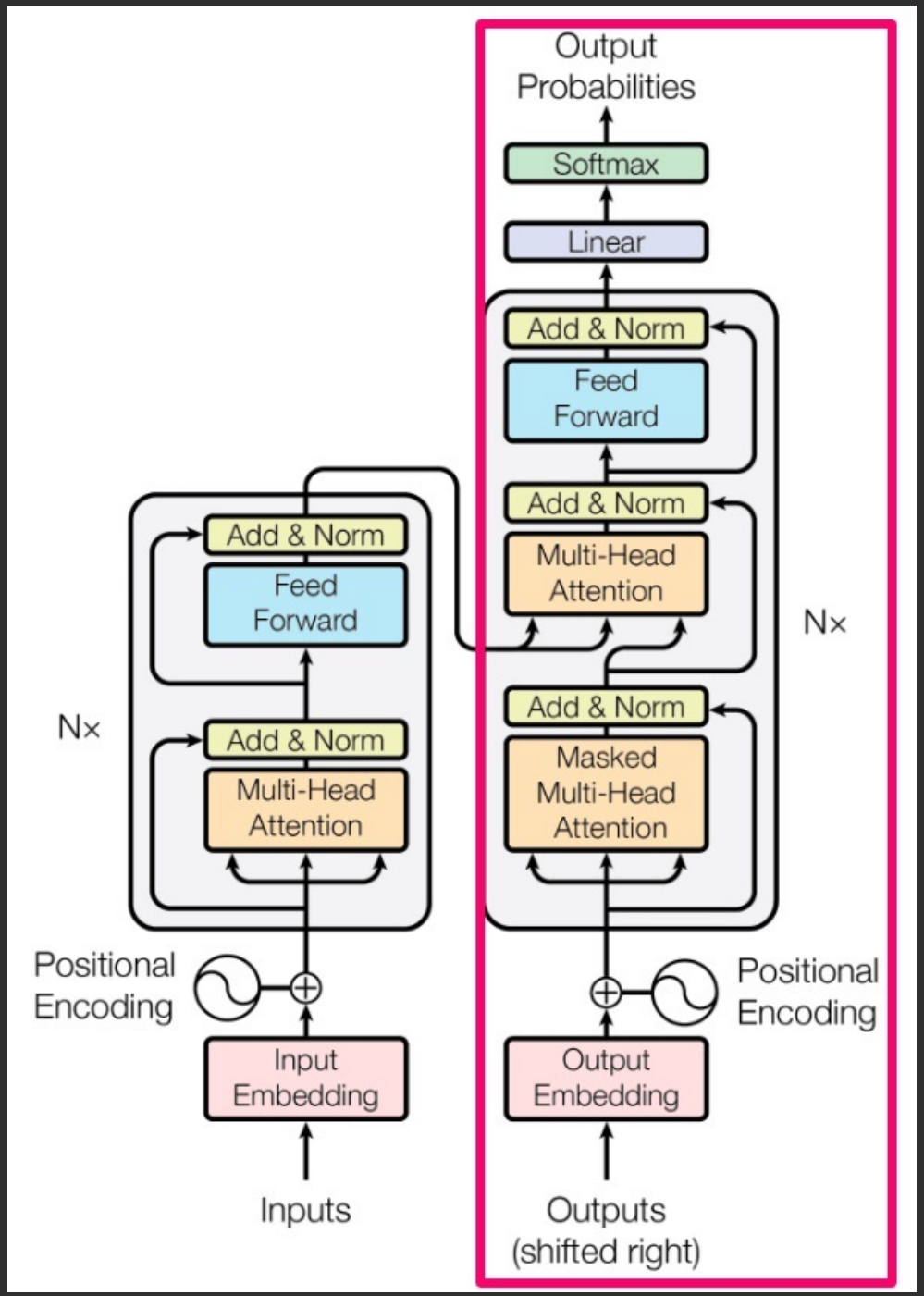

Encoder
4 token
I love NLP
3
Decoder
8 token

5CV

English → "I will go to market"

Hindi → "Main market jaaunga"

3CV



Cross Attention

Encoder → K V → CV

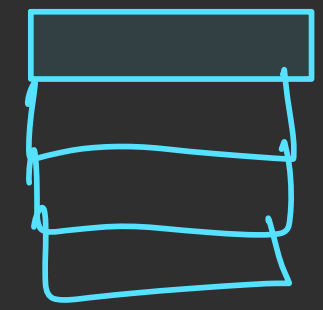
Decoder → Q → CV

We are friends

dost

<S> Hum dost Hain

3 CVs

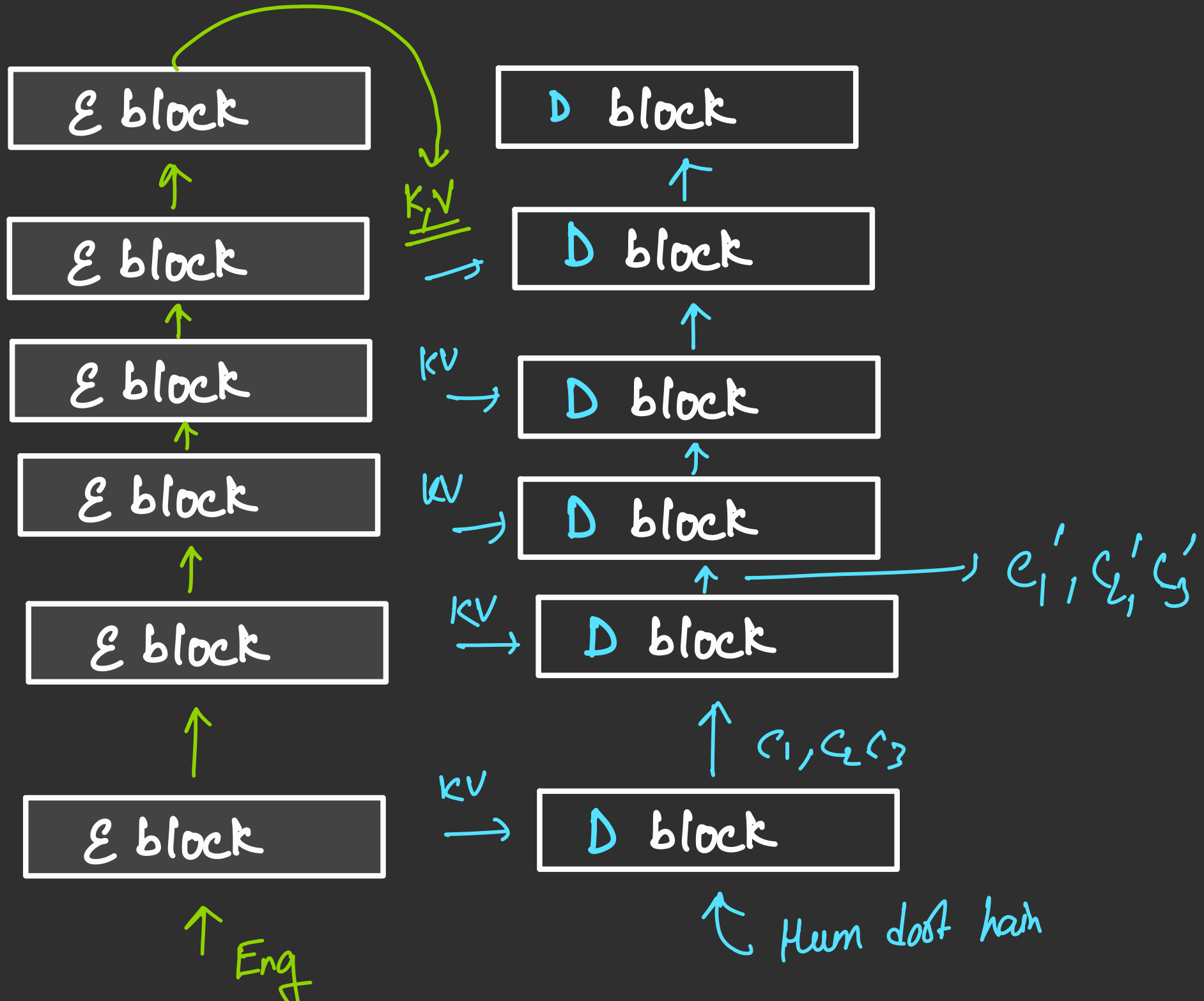


$$Q = QW_Q$$

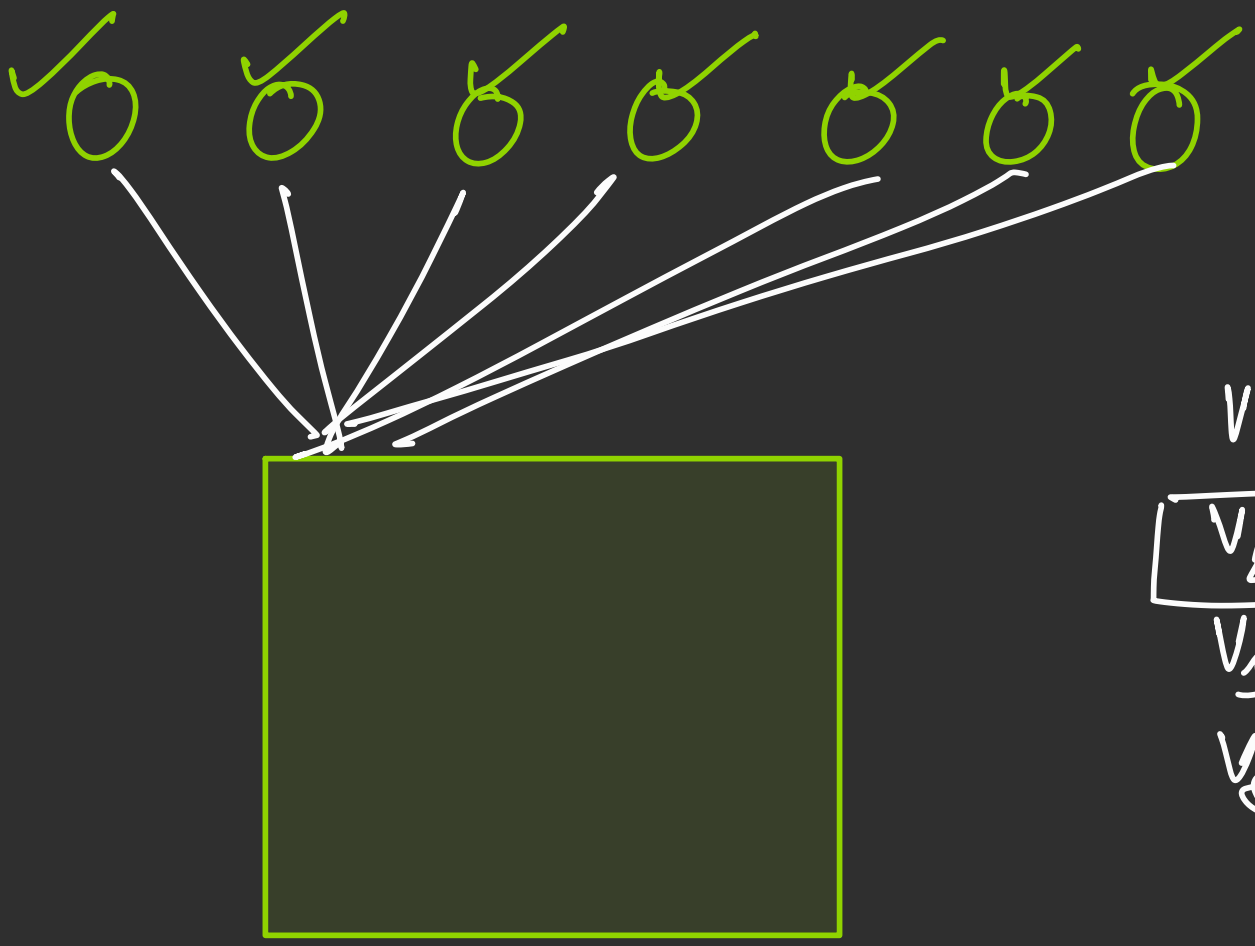
$$K = KW_K$$

$$V = VW_V$$

$$\underbrace{\mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)}\cdot V$$


E block
E block
E block
E block
E block
E block
D block
D block
D block
D block
D block
D block
K,V
K,V
K,V
K,V
K,V
c1, c2 c3
c1' c2' c3'
Eng
Hum dost hain

(softmax)



Vocab collection

V_1 - 0.02

V_2 - 0.98

V_3 - 0.001

V_4 - 0.01

Hum doot hain 0.96

Hum doot 0.96

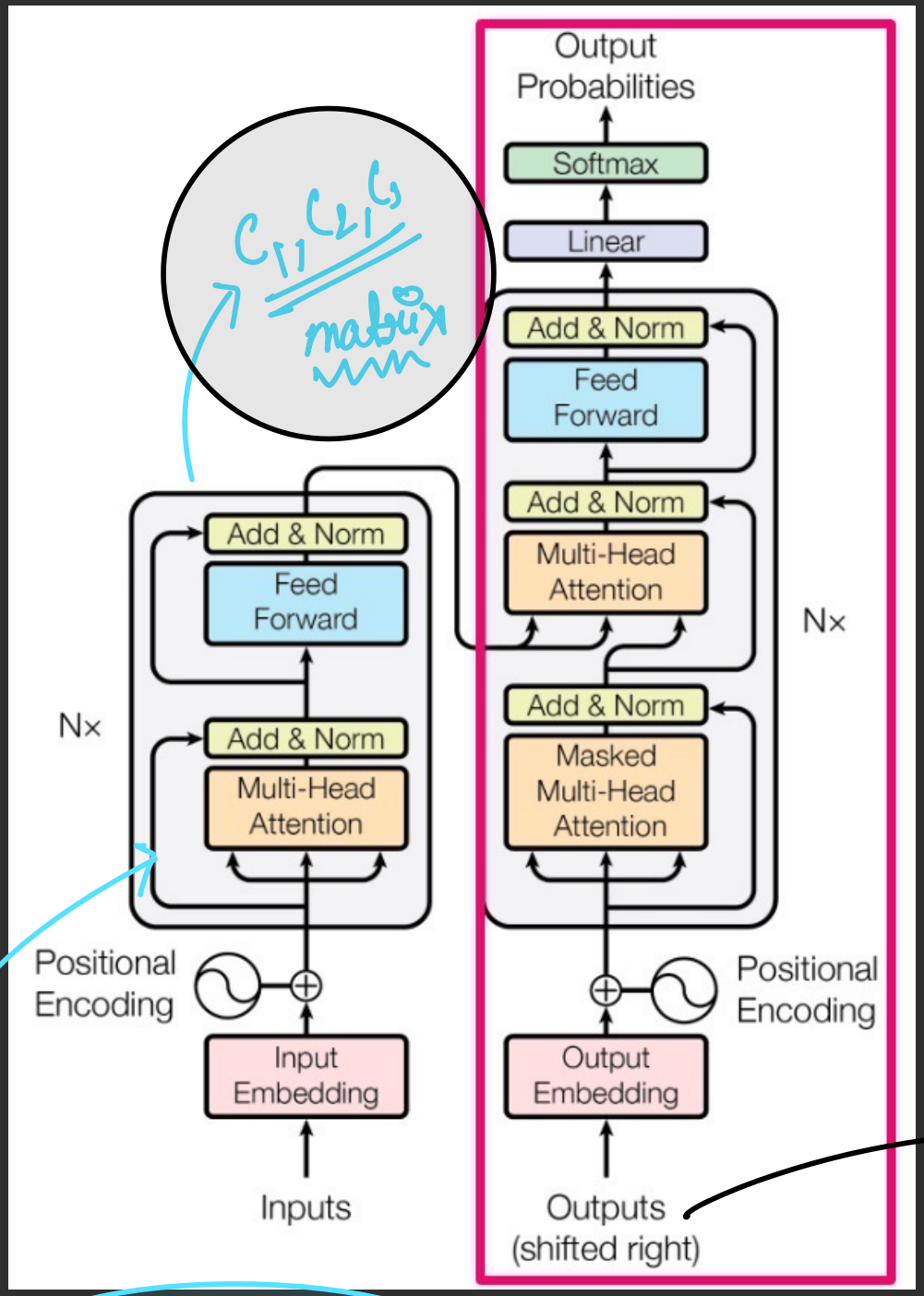
↳ n option

date	values
1	36
2	24
3	36
4	33

100, 102, 101, 150, 103

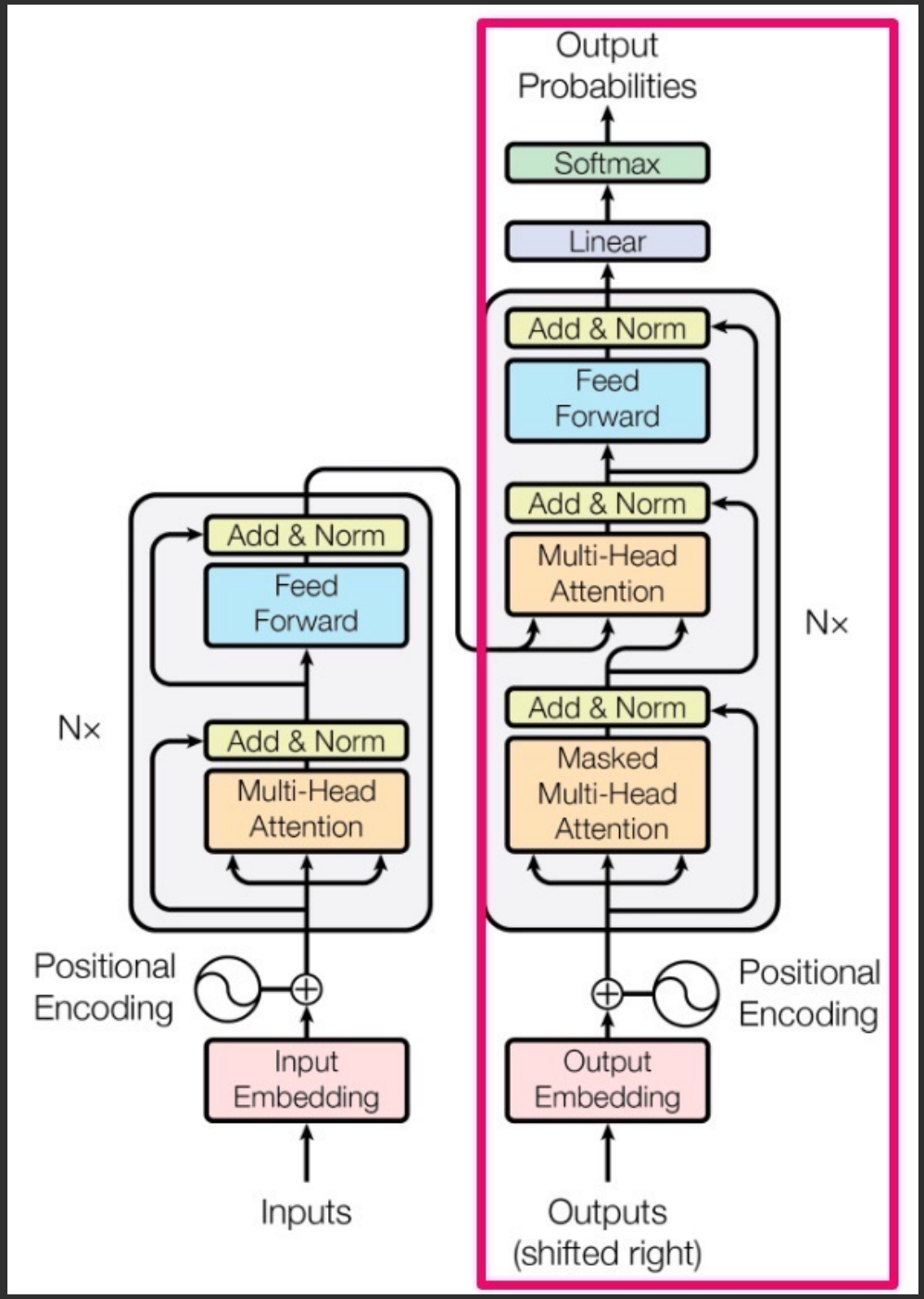
unsuall No

self Attent



We are friends

Hum dost Hain



GPT → Decoder only

BERT → Encoder

$5x = 10$

$x = 10/5$

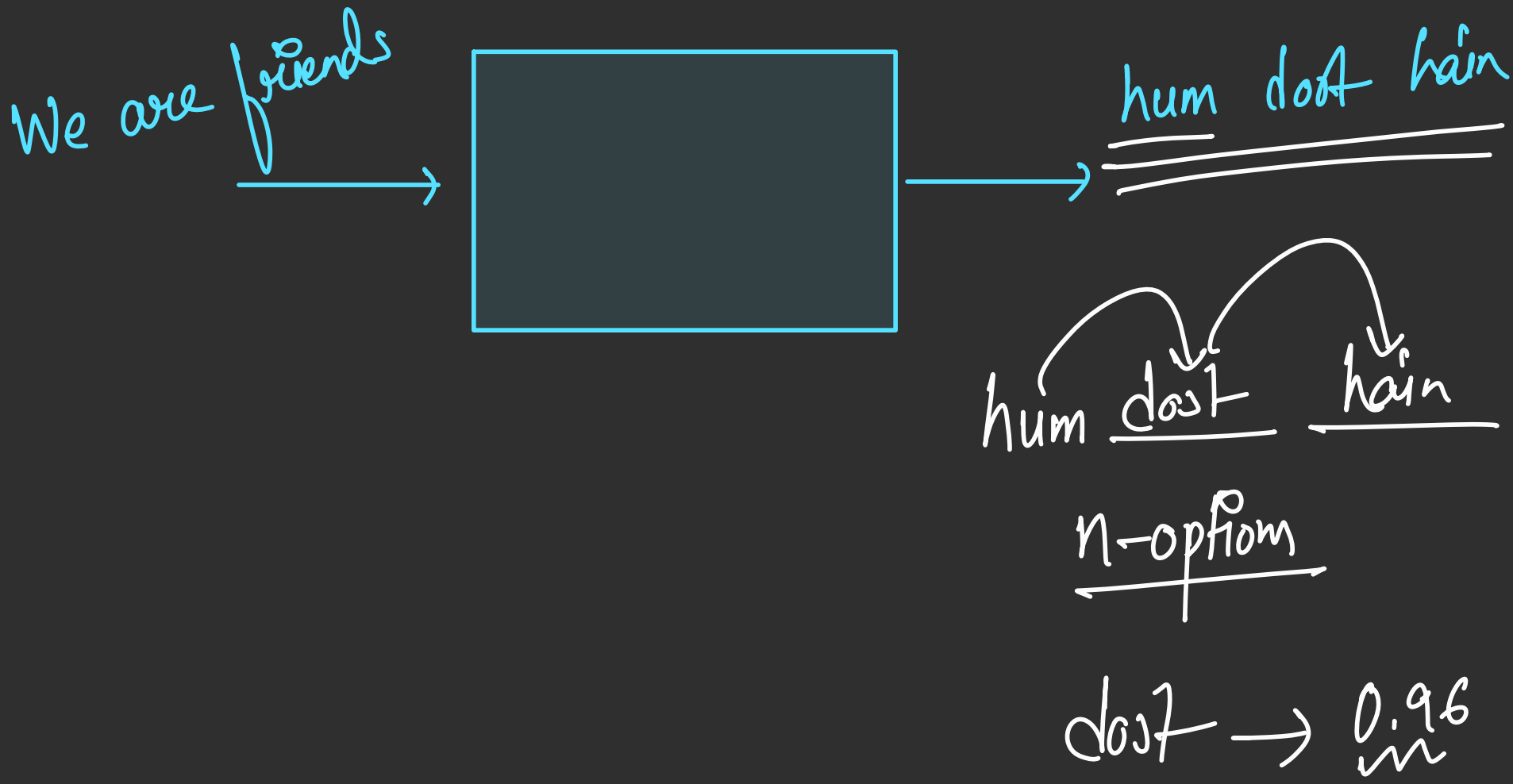
$x = 2$

$10 \times 2 = 20$

$$K = \underline{K} \cdot W_K$$

$$Q = \underline{Q} W_Q$$

$$V = \underline{V} \cdot W_K$$


We are friends
hum dost hain
hum dost hain
n-options
dost → 0.96